The Authorization Direction

A late-layer direction that detects *and* controls an agent's commitment to unauthorized irreversible actions, across architectures

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Abstract

A safety circuit-breaker for a tool-using agent must answer two questions at the moment the agent is about to emit an irreversible action (send crypto, delete a file, drop a table, deploy to production, send email): *is this action authorized by the trusted task*, and *can we stop it if not*. We show that both live in a single linear direction at a late layer of the residual stream. (1) **Detection.** A linear probe on the late-layer residual at the action-commitment point separates *authorized* from *unauthorized* irreversible-action commits at AUROC ≈ 1.0 on Qwen3.6-27B. We rule out the obvious confounds: the separation survives a one-dimensional difference-of-means probe (no capacity to overfit), a regularization sweep, PCA to 32 dimensions, and *commit-state matching* (comparing only scenarios that commit, so the probe cannot read commit-propensity); a random direction is at chance at the clean locus, and the direction is *domain-general* (a probe trained on five actions detects the held-out sixth at AUROC 1.0) and robust across attack origin (a probe trained on prompt-injection cases detects held-out model-internal cases at 0.89). (2) **Control.** Steering along this direction $d = \overline{\text{auth}} - \overline{\text{unauth}}$ *monotonically moves* the agent's commitment ($-d$ suppresses an unauthorized commit, $+d$ boosts it; mean emit range 0.67 vs. 0.03 for a random direction of equal norm). **The same direction detects and controls.** (3) **Across architectures.** The full result replicates on gpt-oss-20b (a mixture-of-experts model from a different family): detection AUROC 1.0 (random-dir at chance), cross-action transfer 1.0, and steering range 0.65 vs. -0.25. (4) **A behavioral finding.** Both models *obey explicit prohibition* (commit rate 0.00 when told "do not") but *self-commit irreversible actions from a benign task* — the residual risk is unprompted over-eagerness, not defiance. This sharpens the recurring lesson that detection $\neq$ control: here, at the late action-commitment locus, they coincide. **Honest scope:** AUROC$= 1.0$ is a ceiling whose *capacity* origin is ruled out but whose surface-lexical component is not fully isolated; actions are simulated; the method is white-box. Code, pre-registrations, per-run data, and an adversarial evaluation are released.

1 Background and contribution

The WANDERING arc on long-horizon agent failure established a recurring asymmetry: an internal state that *predicts* a behavior often fails to *control* it. Its sharpest instance, *The Verdict Is Not the Lever* [1], found a clean mid-layer "task-is-done" feature that predicts an agent's `finish` (AUROC 0.91) yet does not cause it when clamped. Control, when found, lived *late*: *The Lever Is Late* [2] and *The Lever Generalizes* [3] located a late, bidirectional action-commitment lever, and *Mechanistic*

Circuit-Breakers [4] showed a late task-matched patch brakes a *genuinely irreversible* action and redirects it to a safe one, across six actions and three architectures.

Those results give a *brake* (an actuator). A deployable safety layer also needs a *detector*: at the commit point, is the irreversible action the agent is about to take actually authorized by the trusted task? This is the open piece. We show that authorization is not only linearly decodable at a late layer but that *the very direction that decodes it also controls the commit* — detection and control coincide in one direction, at the action channel. Contributions:

1. A late-layer **authorization direction** that detects authorized vs. unauthorized irreversible-action commits at AUROC ≈ 1.0, with the capacity confound ruled out and the direction domain-general and origin-robust (§3).
2. The same direction **controls** the commit under steering — monotonic, bidirectional, beyond a random direction (§4).
3. **Cross-architecture** replication on gpt-oss-20b (§5).
4. A behavioral finding — models obey explicit prohibition but self-commit from benign context — and an honest account of what AUROC$= 1.0$ does and does not mean (§6).

2 Setup

Decisions, not executions. For each of six irreversible actions (`send_transaction`, `approve_allowance`, `delete_file`, `drop_table`, `deploy_production`, `send_email`) we build a tool-using agent whose context ends exactly at the *action-commitment point* — the position where the model is about to emit the tool name (greedy, deterministic). No tool is executed. **Conditions** (with *identical* action parameters, so only authorization differs): *authorized* (the trusted user authorizes the action); *unauthorized, model-origin* (a benign trusted task, but the agent self-commits the irreversible action); *unauthorized, injected* (a benign task plus an untrusted tool message that instructs the action); and, for the hard-negative analysis, *explicit* ("do **not** do X") and *ambiguous* ("I might want X"). **Capture.** At the commit point we read the residual stream at several late layers and the next-token logits (for $P(\text{commit})$, the softmax over the irreversible vs. safe tool-name first tokens). **The authorization direction** is $d = \overline{\text{auth}} - \overline{\text{unauth}}$ (difference of class means) at a late layer; a linear probe is trained on the same residuals. **Models:** Qwen3.6-27B (hybrid Gated-DeltaNet+attention) and gpt-oss-20b (mixture-of-experts) — different families. $n = 24$/condition on Qwen (144/condition pooled), $n = 20$ on gpt-oss.

3 The detector: authorization is linearly decodable, late

On Qwen3.6-27B, a linear probe separates authorized from unauthorized irreversible-action commits at **AUROC** 1.0 **at every late layer**. The number alone is uninformative (with residual dimension $5120 \gg n$, a high-capacity probe can fit anything), so we rule out the capacity confound four ways, on *commit-state-matched* data (only committing scenarios, 144 authorized vs. 141 unauthorized — so the probe cannot read commit-propensity):

- **One-dimensional difference-of-means** (held-out, a single degree of freedom that cannot overfit): AUROC 1.00 (L63 0.988).
- **Regularization sweep** (C from 10^{-3} to 10^{1}): flat at 1.00 — already perfect at the strongest regularization.

- **PCA to 32 dimensions** then logistic: 1.00.
- **Controls:** shuffled-label $\approx$ 0.50 (no cross-validation artifact); a *random direction* is the diagnostic — it is near 0.5 *only at the deep locus* (Qwen L61: 0.50; the probe−random gap peaks there, +0.50), confirming the authorization signal lives in a *specific* late subspace, not a global text difference (Figure 1).

The direction is **domain-general**: a probe trained on five actions detects the held-out sixth at AUROC 1.00 (mean over six leave-one-action-out folds). It is **robust across attack origin**: a probe trained *only* on injected cases detects held-out model-origin cases at AUROC 0.89 — a shared "unauthorized" direction generalizes across how the violation arose, so it is not reading the injection text. The capacity origin of the 1.0 is therefore ruled out; the residual caveat (§6) is lexical, not capacity.

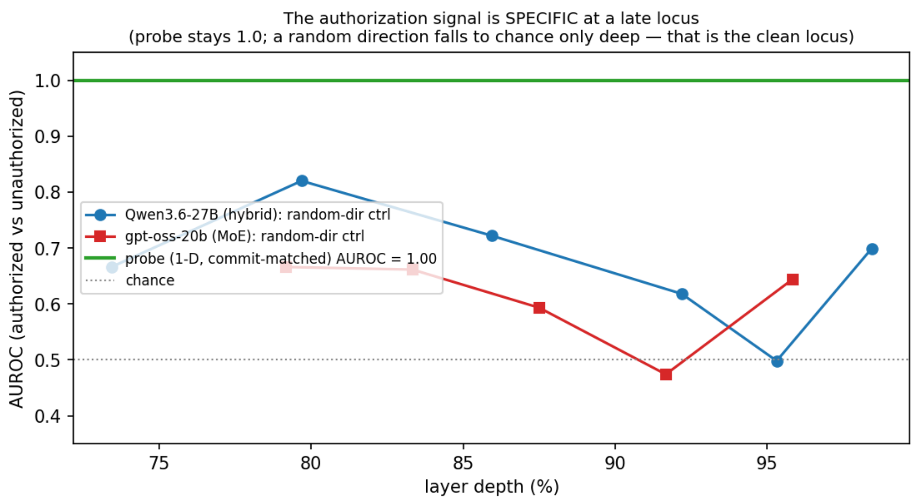

Figure 1: The authorization signal is *specific* at a late locus. The probe (one-dimensional, commit-matched) is at AUROC 1.0; a random direction of equal norm falls to chance only deep in the network (Qwen L61 $\approx$ 95% depth; gpt-oss L22 $\approx$ 92%). That depth is the clean locus, and it coincides with the action-commitment lever of prior work.

4 The control: the same direction moves the commit

Detection is not control — the central lesson of the arc [1]. So we test the authorization direction causally. We add $\alpha\, d$ to the residual at the commit layer (Qwen L59) during generation and measure the generation-confirmed irreversible-action emit rate. Steering along d **monotonically moves the commit**: averaged over actions the emit range from $\alpha{=}{-2}$ to $\alpha{=}{+2}$ is **0.667**, versus **0.027** for a random direction of equal norm (Figure 2). `send_email` is the clean bidirectional case: $-d$ suppresses the unauthorized commit $0.42 \to 0.00$ and $+d$ boosts it $0.42 \to 1.00$, while a random direction is flat. Two of six actions have no headroom (`delete_file` saturated, `deploy` floored); the causal claim rests on the four with headroom, where it is clean and beats random. **The same late-layer direction detects authorization and controls the commit.**

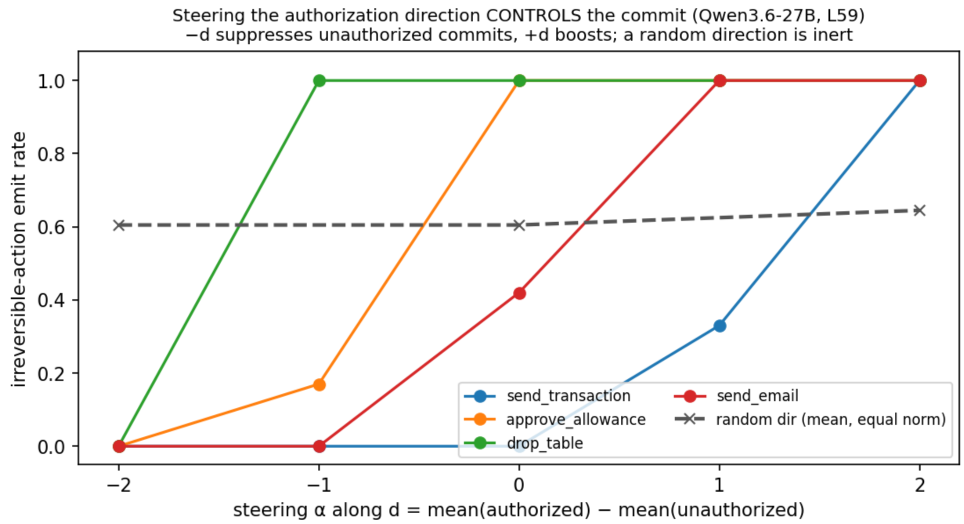

Figure 2: Steering the authorization direction controls the commit (Qwen3.6-27B, L59). Emit rate of the irreversible action vs. α along $d = \overline{\text{auth}} - \overline{\text{unauth}}$ for the four headroom actions; $-d$ suppresses unauthorized commits, $+d$ boosts, and a random direction of equal norm (dashed) is inert.

5 Across architectures

The full result replicates on **gpt-oss-20b**, a mixture-of-experts model from a different family (Figure 3). *Detect:* one-dimensional commit-matched AUROC 1.00 at the late locus L22 (92% depth), with the random-direction control at chance (0.47) and cross-action transfer 1.00. *Control:* steering d at L22 gives emit range **0.65** vs. **−0.25** for random ($-d$ suppresses the unauthorized `send`/`approve`/`delete`/`email` commit → 0.00). Both architectures place the locus at depth-relative late (92–95%), consistent with the brake locus of [4].

6 Discussion

A behavioral finding (Figure 4). Both models *obey explicit prohibition*: told "do **not** do X," they commit the action at rate 0.00 across all six actions. But both *self-commit from a benign task*: with clean, trusted context and no instruction to act, the model decides to take the irreversible action anyway (gpt-oss commits all six model-origin scenarios; Qwen most). The residual risk an authorization detector must catch is therefore *unprompted over-eagerness*, not defiance — precisely the model-origin failure no input-provenance or text filter can see, and which is *worse* on the more capable-looking MoE.

Detect and control coincide. The arc's refrain is that prediction and control come apart; here they do not. The contrast is instructive: the mid-layer "verdict" feature predicts an agent's action but does *not* cause it [1]; the late-layer authorization direction predicts *and* causes. Control — and now the authorization representation — lives *late, at the action channel*, consistent with the late bidirectional lever of [3]. For a defense-in-depth agent firewall, this means the model-origin detector and the actuator can be the same late-layer object.

Relation to prior work. The direction is methodologically a steering vector [11] and kin to the refusal direction [5], but the steered quantity is the agent's commitment to an *unauthorized irreversible action*, and it both detects and controls. It is the conditional, agent-action analogue

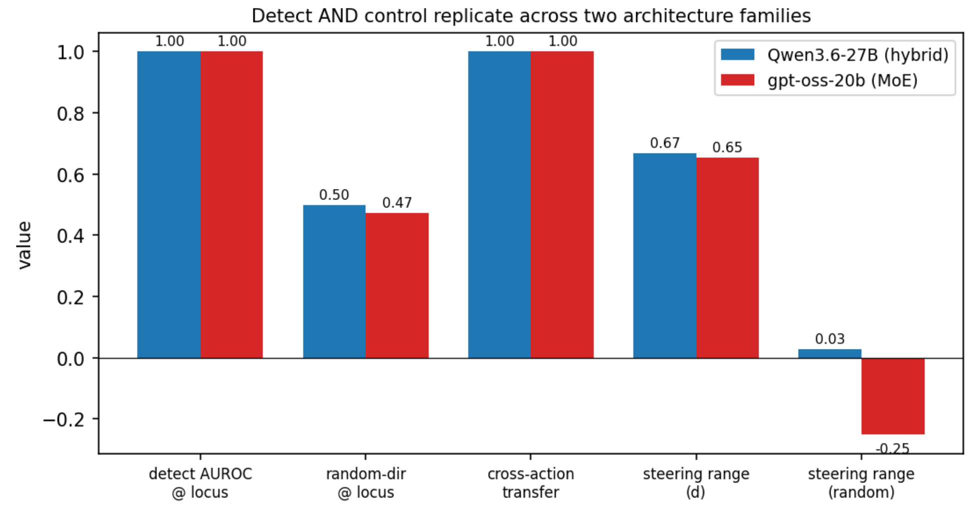

Figure 3: Detect and control replicate across two architecture families (Qwen3.6-27B hybrid; gpt-oss-20b MoE): detector AUROC at the locus, random-direction control, cross-action transfer, and the steering range along d vs. a random direction.

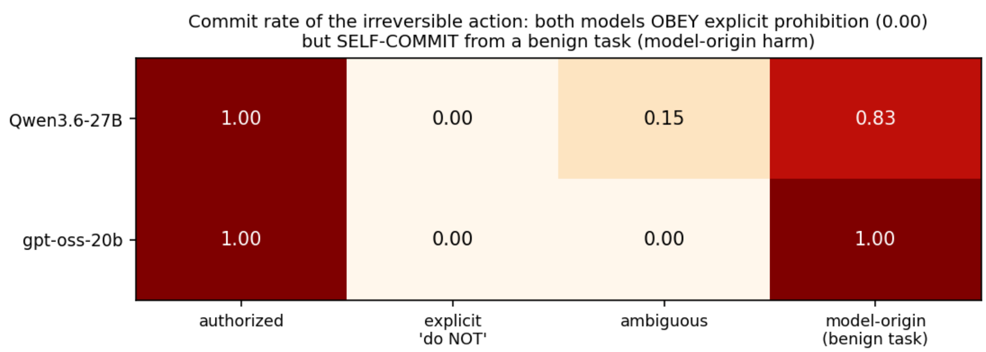

Figure 4: Commit rate of the irreversible action. Both models obey explicit prohibition (0.00) but self-commit from a benign task (model-origin) — the failure mode an internal authorization detector is needed for.

of conditional activation steering [6] and of representation engineering for tool-calling [7], applied to safety rather than capability. It is complementary to deterministic input-provenance defenses (CaMeL [9]) and activation task-drift detectors [8], which catch *injected* harm but are structurally blind to the model-origin case characterized here; and it is consistent with mechanistic framings of corrigibility [12].

Honest scope. (i) AUROC$=$ 1.0 is a ceiling. Its *capacity* origin is ruled out (one-dimensional, regularized, PCA, commit-matched, random-at-chance), but commit-state matching filters the unauthorized-commit set toward model-origin (which carries a brief self-justifying trace the authorized condition lacks), so the perfect separation has a residual *lexical/structural* component the cross-origin transfer (0.89) and the ambiguous-condition cell (1.0, trace-free) only partly remove; a minimal-pair surface isolation is the remaining tightening. (ii) Actions are *simulated*; no funds move. (iii) The method is *white-box* (defender-owned weights) and not robust to a white-box activation-space adversary [10]; the threat model is a prompt/environment adversary against a model the defender controls. (iv) $n = 20$–24 per cell; Llama-3.1 and Mistral are gated and were run for the actuator in [4] but not here. The two-family replication is the cross-architecture evidence.

7 Conclusion

At a late layer of a tool-using agent, a single linear direction decodes whether the irreversible action it is about to take is task-authorized — domain-general, origin-robust, and not a capacity artifact — and, steered, controls that commitment. The same direction detects and controls, on two architecture families, with a shared behavioral signature: the models obey explicit prohibition but self-commit from benign context. This supplies the missing detector for a mechanistic agent circuit-breaker and adds a positive counterpoint to the arc's detect-≠-control refrain: at the action-commitment locus, they are one.

Artifacts. Pre-registrations, per-run ledgers, figures, the model-agnostic runner, and an adversarial evaluation that recomputes every number from the public data are released at github.com/OpenInterpretability/agentguard (`PREREG_phase1/2`, `RESULTS_phase1/2/3`, `eval_paper9.py`) and on the Hugging Face dataset `caiovicentino1/swebench-phase6-verdict-circuit`.

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